



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
FULL SYLLABUS
04-05-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 195

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

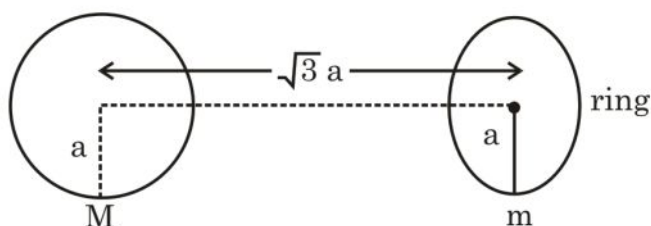
Negative Marks : -1 In all other cases

- Free ^{238}U nuclei kept in a train emit alpha particles. When the train is stationary and a uranium nucleus decays, a passenger measures that the separation between the alpha particle and the recoiling nucleus becomes x in time t after the decay. If a decay takes place when the train is moving at a uniform speed v , the distance between the alpha particle and the recoiling nucleus at a time t after the decay, as measured by the passenger will be :-
 (A) $x + vt$
 (B) $x - vt$
 (C) x
 (D) depends on the direction of the train.
- Two identical spherical masses are kept at some distance as shown. Potential energy when a mass m is taken from surface of one sphere to the other.



- Increases continuously.
- Decreases continuously.
- First increase then decreases.
- First decreases then increases.

3. A uniform ring of mass m and radius a and a sphere of mass M and radius a are released in free space. Consider only gravitational force between sphere and ring. Find acceleration of ring at given separation.



- (A) $\frac{\sqrt{3}GM}{8a^3}$
- (B) $\frac{\sqrt{3}GM}{4a^2}$
- (C) $\frac{\sqrt{3}GM}{8a^2}$
- (D) $\frac{GM}{\sqrt{3}a^2}$
4. A car drives along a road that has small depressions regularly spaced about 8.0m apart. When four 80 kg passengers enter the car, it sinks down by 1.6 cm. At approximately what speed the car might travel so that travelling in the vehicle becomes very uncomfortable?
- (A) 5π m/s
- (B) 10π m/s
- (C) 2π m/s
- (D) 20π m/s
5. Three tuning forks in the audible range produce beats: B and C produce beats at 7 Hz, A and C produce beats at 8 Hz. The frequency of B is 6% higher than that of A.
- (A) Frequency of tuning fork A is 265 Hz.
- (B) Frequency of tuning fork B is 265 Hz.
- (C) Frequency of tuning fork C is 250 Hz.
- (D) None of the above

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

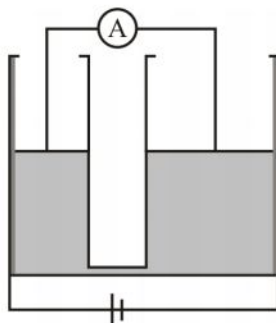
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. Two identical thin homogeneous nichrome wires are mounted into the side non-conducting walls of vertical communicating vessels (vessel cross-sectional area $S = 100 \text{ cm}^2$ and $3S$). The length of the wires is equal to the height of the vessels. An ideal battery with a voltage at the terminals $U = 46.2 \text{ V}$ is connected to the lower ends of the wires. The vessels are filled upto a level 50 cm and covered with light and thin conductive pistons. The pistons do not allow water to pass through, have contact with the wires and can move without friction inside the vessels without leaving them, thanks to stoppers.

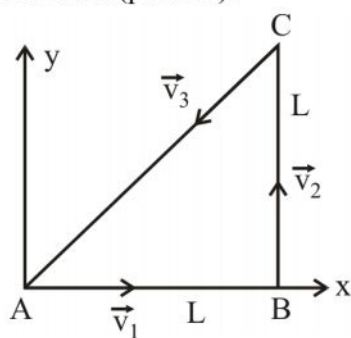
An ammeter is connected to the pistons using flexible wires (the length of the wires allows the pistons to move freely). The installation diagram is shown in the figure. When a non-conducting weight weighing 1 kg is placed on a small piston, the ammeter shows a current strength of $I_1 = 2.31 \text{ A}$. If 3 such weights are placed on a large piston, the ammeter shows a current strength of $I_2 = 2.1 \text{ A}$.

We can assume that the connecting tube has a negligibly small cross-section. The conductivity of water is also very low. Water density $\rho = 1000 \text{ kg/m}^3$.

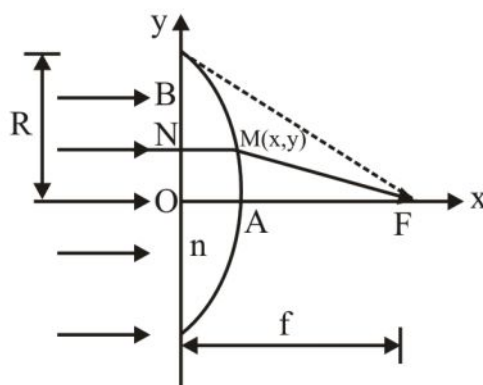


- (A) The resistance of the ammeter R_a is 1 ohm .
 (B) The reading of ammeter when no weight is placed on the pistons is approx. 2.2 A .
 (C) The maximum current through the ammeter is approx. 3.22 A .
 (D) The minimum current through the ammeter is approx. 1.5 A .

7. Three tigers simultaneously begin to move along a horizontal surface with constant speeds. The velocity of the first tiger at any moment of time is directed towards the second, the velocity of the second towards the third, and the velocity of the third towards the first. At the initial moment of time, tigers form a right triangle with legs equal to L (Figure). Consider the size of the tigers to be much smaller than L . The speed of the first tiger is v , and the speeds of the second and third tigers v_2 and v_3 are such that during the movement the angles in the triangle ABC formed by the tigers remain constant. Let's introduce a coordinate system as shown in the figure. The origin of coordinates coincides with the position of the first tiger at the moment of start (point A).



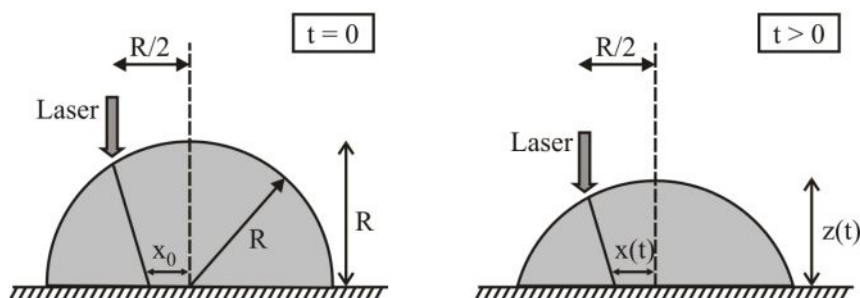
- (A) The time after which the tigers will meet is $\frac{L}{v}$.
- (B) The speed of second tiger v_2 is $\frac{v}{2}$.
- (C) The speed of third tiger v_3 is $\frac{v}{2}$.
- (D) The coordinates of the point where the tigers will meet is $\left(\frac{4L}{5}, \frac{L}{5}\right)$
8. As shown in figure, a plano-convex lens has a refractive index n and is placed in the air. The radius of the lens aperture is R . Take a point F on the principal optical axis of the lens, $OF = f$. When parallel light is incident along the principal optical axis, all the light rays converge at point F .



- (A) The shape of the convex surface of the lens should be hyperbola.
- (B) The shape of the convex surface of the lens should be parabola.
- (C) The maximum value of aperture R of the lens is nf .
- (D) The maximum value of aperture R of the lens is $\sqrt{n^2 - 1} f$.

9. An ice hemisphere with radius R and refractive index n lies on a warm flat table and melts slowly. The rate of heat transfer from the table to the ice is proportional to the area of contact between them. It is known that the ice hemisphere completely melts in time T_0 . Throughout the process, a laser beam is incident on the ice from above. The beam is vertically incident at a distance of $R/2$ from the axis of symmetry (see figure).

Assume that the temperature of the ice and the surrounding atmosphere are 0°C and remains constant during the melting process. The laser beam does not transfer energy to the ice. The melting water immediately flows off the table, and the ice does not move along the table.



- (A) The value of x_0 (point on the table where the laser beam strikes at $t = 0$) is $\frac{2R}{(\sqrt{12n^2 - 3} + 1)}$.
- (B) The value of x_0 (point on the table where the laser beam strikes at $t = 0$) is $\frac{2R}{(\sqrt{4n^2 - 3} + 1)}$.
- (C) The height of ice z at any time t is $z = R \left(1 - \frac{t^2}{T_0^2} \right)$.
- (D) The time at which the value of x (point on the table where the laser beam strikes) becomes $\frac{R}{2}$ is $\frac{\sqrt{3}}{2} T_0$.

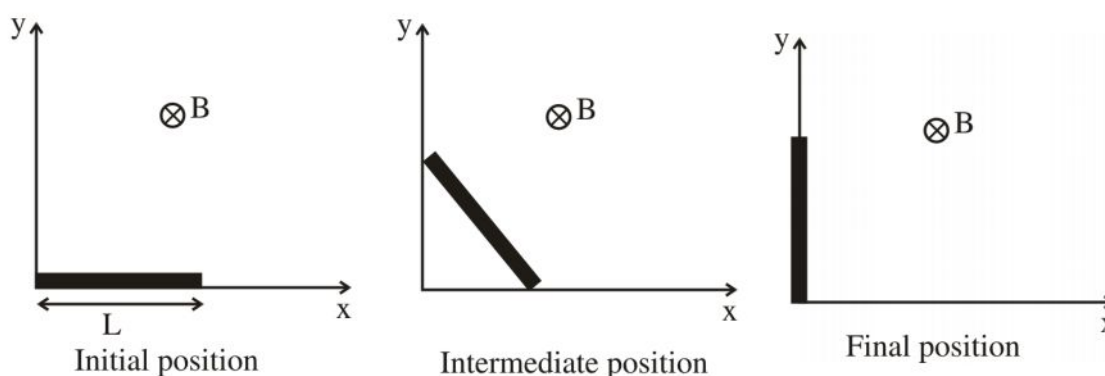
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

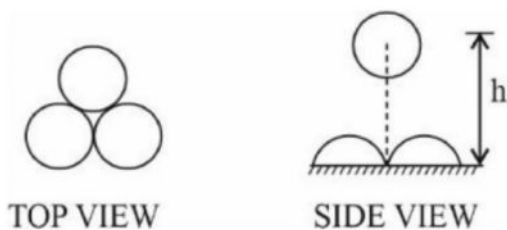
Zero Marks : 0 In all other cases

1. A metallic rod of mass m and length L can slide without friction on two perpendicular wires. Entire arrangement is located in the horizontal plane. A magnetic field of magnitude B exists perpendicular to this plane in the downward direction. The wire have negligible resistance compared to the rod whose resistance is R . Initially rod is along one of the wire so that one end of it is at the junction of two wires show the rod is given an initial angular speed ω such that it slide with two ends in contact with the two wires & just comes to rest in an aligned position with the other wire. The value of ω is $\frac{3\pi B^2 L^2}{\alpha m R}$. The value of α is :- (Neglect self inductance of system.)

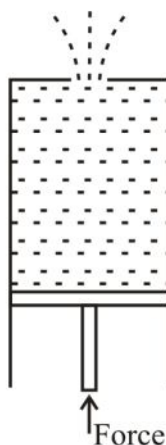


2. Two blocks of metal with heat capacities $C_1 = 3\text{J/K}$ and $C_2 = 6\text{ J/k}$, are interconnected by a rod of length ℓ and cross-sectional area S and fairly low heat conductivity K . The whole system is thermally insulated from the environment. At a moment $t = 0$ the temperature difference between the two blocks of metal equals $(\Delta T)_0$. Assuming the heat capacity of the rod to be negligible, if $\frac{\Delta T}{\Delta T_0} = e^{-\frac{SKt}{\alpha t}}$, then α is

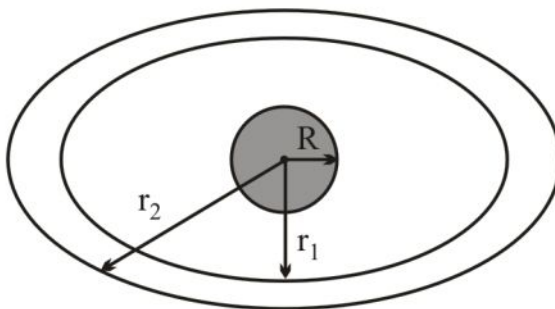
3. A sphere of mass m falls symmetrically on a set of three hemispheres placed in contact with each other on smooth horizontal ground from a height h . Mass of each hemisphere is $m/2$ and radius of each body is R . Neglect any rebounding of hemisphere. If sphere comes to rest after the simultaneous collision with three hemispheres then ratio of KE of sphere before collision and KE of each hemisphere after collision is p . Find the value of p .



4. A syringe is filled with water up to volume 20 cm^3 . The area of cross-section of the cylinder is 5 cm^2 . The syringe is held fixed vertically and its 90 gm piston is pushed upward by external agent with constant speed. A water beam coming out of the small nozzle (hole area 1 mm^2) has speed 2 m/s . Neglecting friction and viscous nature of water, find the work done by the agent (in 10^{-2} J) in fully emptying the syringe. (Take : $g = 10 \text{ m/s}^2$)



5. A positive charge with a constant surface density σ is distributed over the surface of a thin flat non-conducting ring with internal and external radii of r_1 and r_2 , respectively. The center of a non-conducting ball with radius $r < r_1$ coincides with the center of the ring. The sphere is uniformly charged in volume with a positive charge Q . At what values of σ (in C/m^2) is the resulting field strength vector at points on the surface of the ball located in the plane of the ring is zero? (Take : $R = 1\text{cm}$, $r_1 = 2\text{m}$, $r_2 = 3\text{m}$ and $Q = \pi \times 10^{-6} \text{C}$)



6. We use potential energy as mgh which is an approximate value, the actual value is $mgR^2 \left(\frac{1}{R} - \frac{1}{R+h} \right)$. Where R is radius of earth (6400 km) and h is height from earth's surface. Find maximum height (in km) from earth's surface where the potential energy expression has error of less than 5% of true value.

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

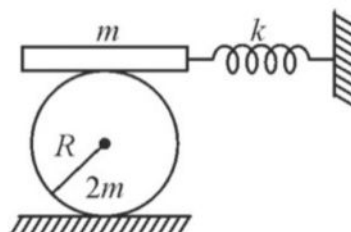
For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 If none of the options is chosen

Negative Marks : -1 In all other cases

1. A uniform plank of mass m , free to move in the horizontal direction, is placed on the top of a solid cylinder of mass $2m$ and radius R . The plank is attached to a fixed wall with the help of a light spring of force constant k . The plank is slightly displaced towards spring and released. There is no slipping anywhere. For any displacement x , amplitude x_0 and velocity v of the plank, match the column-I with column-II and mark the correct option from the codes given below.



	Column-I		Column-II
(A)	Velocity of centre of cylinder.	(P)	$\frac{1}{2}kx_0^2$
(B)	Energy of oscillations of the system.	(Q)	$\frac{v}{2}$
(C)	Magnitude of acceleration of the plank.	(R)	$\frac{7}{8}mv^2 + \frac{1}{2}kx^2$
(D)	Angular frequency of oscillations.	(S)	$\frac{4kx}{7m}$
		(T)	$\sqrt{\frac{4k}{7m}}$

2.

Column –I (Situation)		Column –II (Values in MKS units)	
(A)	Magnitude of electric field at (1,1,1) if potential is given by $V = -\frac{1}{x} - \frac{1}{y} - \frac{1}{z}$, is	(P)	0.4
(B)	The charge flowing through a resistance 6Ω varies with t as $Q = 4t - 2t^2$. The total heat produced in resistance is	(Q)	$\sqrt{3}$
(C)	A milliammeter of range 10mA has a coil of resistance 0.99Ω . To use it as an ammeter of range 1A, the required shunt must have a resistance of	(R)	0.01
(D)	Two unlike point charges of same magnitude 10 mC are placed on x-axis at a separation of 8 cm. If both charges placed symmetrically about origin in external electric field $\vec{E} = 500\text{V m}^{-1} \hat{i}$, then potential energy of the system may be [Assume potential energy to be zero when dipole is perpendicular to electric field]	(S)	32
		(T)	None of these

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Which of the following pair of complex both show stereo isomerism as well as structural isomerism.

(A) $[\text{Pt}(\text{en})(\text{CN})_2]$, $[\text{Pt}(\text{en})_3]\text{Cl}_3$

(B) $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$

(C) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2\text{Br}_2]$, $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Br}$

(D) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$, $[\text{Co}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_2\text{Cl}_2]\text{NO}_3$
2. Which of the following is not disproportionation reaction.

(A) $\text{P}_4 \xrightarrow{\text{NaOH}}$

(B) $\text{HNO}_2 \xrightarrow{\Delta}$

(C) $\text{XeF}_6 \xrightarrow{\text{H}_2\text{O}/\text{OH}^- (\text{excess})}$

(D) $\text{NaOCl} \xrightarrow[\Delta]{\text{dil. HCl}}$
3. The rate of effusion of two gases 'A' and 'B' under identical conditions of temperature and pressure are in the ratio of 2 : 1. What is the ratio of rms velocity of their molecules if temperature of A & B are in ratio 2 : 1

(A) $\sqrt{2} : 1$

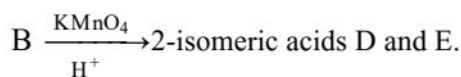
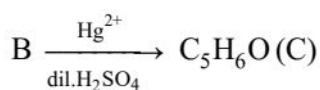
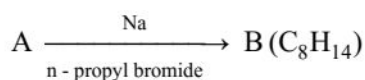
(B) 2 : 1

(C) 1 : $\sqrt{2}$

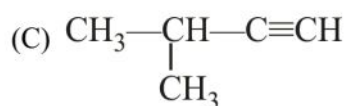
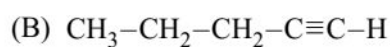
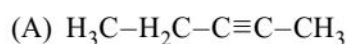
(D) $2\sqrt{2} : 1$

4.

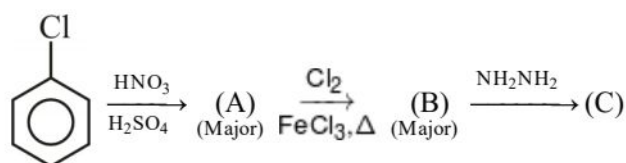
A(Hydrocarbon) ($C \Rightarrow 88.24\%$) [Molecular weight of A = 68]



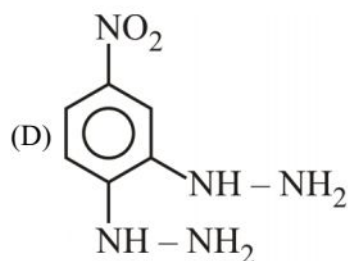
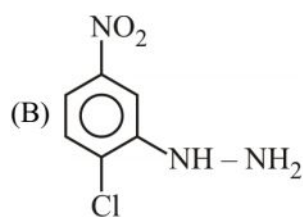
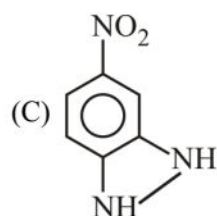
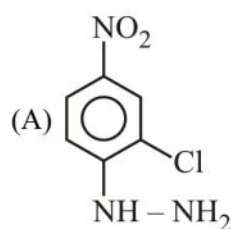
'A' is :



5.



(C) is



SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. Which of the following order is/are correct?

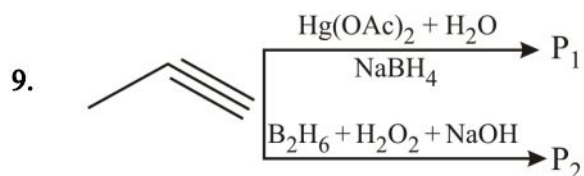
- (A) $\text{RbCl} > \text{LiCl}$ (Ionic character)
- (B) $\text{AlCl}_3 > \text{BCl}_3$ (Lewis acidic strength)
- (C) Atomic nitrogen > molecular nitrogen (Ionization energy)
- (D) $\text{Li}_2\text{CO}_3 > \text{Cs}_2\text{CO}_3$ (Thermal stability)

7. In the homolytic cleavage of N_2O_4 which of the following property(ies) changes.

- (A) Polarity
- (B) Bond length of NO
- (C) Bond angle ($\text{O} - \hat{\text{N}} - \text{O}$)
- (D) Magnetic behaviour

8. Which of the following is/are correct at 298K :

- (A) $\Delta_f G^\circ (\text{H}_{2(g)}) = 0$
- (B) $\Delta_f G^\circ (\text{H}_2\text{O}, \ell) = 0$
- (C) $\Delta_f G^\circ (\text{D}_2, \text{g}) = 0$
- (D) $\Delta_f G^\circ (\text{H}^+, \text{aq.}) = 0$



P_1 & P_2 will be distinguished by

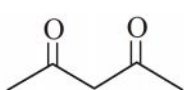
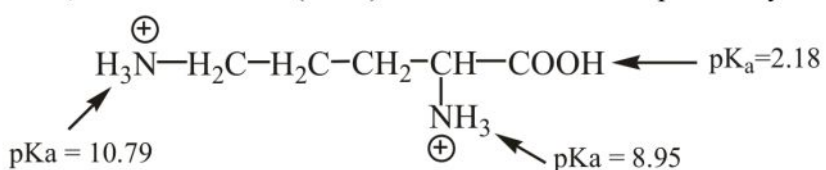
- (A) 2, 4, D.N.P
- (B) alkaline Cu^{2+} (tartarate)
- (C) NaHSO_3
- (D) Schiff's reagent

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 In all other cases

- Find the sum of element which does not have f-subshell e^- .
Th, La, Hg, Pb, Os, Pt, Rh
- When CrI_3 reacts with H_2O_2 in presence of NaOH forming Na_2CrO_4 , NaIO_4 & H_2O then find how many times will be rate of disappearance of NaOH compared to CrI_3 .
-  Number of iodoform molecules produced per molecule.
- If pK_a values for protonated Lysine molecule are $\text{pK}_{a1} = 2.18$; $\text{pK}_{a2} = 8.95$; $\text{pK}_{a3} = 10.79$ respectively then, Calculate value of $(x - 1)$ where x is iso-electric point of lysine.

Answer should be nearest integral value
- Identify reactions in which new C—C bond is formed ?
(i) Aldol (ii) Cannizaro (iii) Reimer Tiemann
(iv) Hoffmann's Bromamide (v) Sandmeyer reaction
(vi) Gattermann Aldehyde synthesis (vii) Friedel craft reaction
(viii) Wurtz reaction
- Find the maximum number of electron(s) in an orbital which having $n = 4$ and $\ell = 2$.

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **Column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- For each entry in **Column-I**, darken the options of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then chosen these three options. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

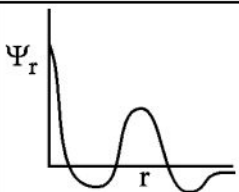
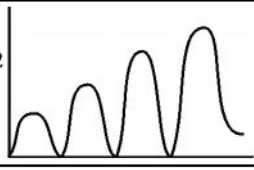
Zero Marks : 0 If none of the options is chosen

Negative Marks : -1 In all other cases

1. Match the column :

Column-I		Column-II	
(A)	Tetragonal and Hexagonal	(P)	are two crystal systems
(B)	Cubic and Rhombohedral	(Q)	$a = b \neq c$
(C)	Monoclinic and Triclinic	(R)	$a \neq b \neq c$
(D)	Cubic and Hexagonal	(S)	$a = b = c$
		(T)	$\alpha = \beta = \gamma$

2. Column-I and Column-II contain data on Schrodinger Wave – Mechanical model, where symbols have their usual meaning. Match the columns :

Column-I		Column-II	
(A)		(P)	4s
(B)		(Q)	5p _x
(C)	$\Psi(\theta, \phi) = K$ (independent of θ & ϕ)	(R)	3s
(D)	Atleast one angular node is present	(S)	6d _{xy}
		(T)	2p _z

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- The value of $\lim_{n \rightarrow \infty} \sum_{n=1}^n \frac{n^2 \tan^{-1} \left(\frac{2n+1}{1+(n^2+n)^2} \right) + (2n+1) \tan^{-1}(n+1)^2}{(n^2+n)^2 \tan^{-1} n^2 \cdot \tan^{-1}(n+1)^2}$ is equal to -
 (A) $-\frac{\pi}{4}$ (B) $\frac{4}{\pi}$
 (C) $\frac{\pi}{2}$ (D) $\frac{2}{\pi}$
- Let $A = [a_{ij}]_3$ be an orthogonal matrix, B and C are two other matrices of same order as A of which B is symmetric and C is skew symmetric such that $A(B+C) = KI$ where K is non zero scalar. If $\text{Tr.}(B) = 8 \text{Tr.}(A) \neq 0$ then K is -
 [Note : $\text{Tr.}(P)$ denotes trace of matrix P]
 (A) 2 (B) 3
 (C) 4 (D) 8
- If one number is selected from each of the three sets $A = \{1,2,3,4,5,6,7\}$, $B = \{1,2,3,4\}$ and $C = \{2,3,4,5,6,7\}$ then probability that product of the number selected from one and sum of the numbers selected from the other two is a prime, is -
 (A) $\frac{1}{7}$
 (B) $\frac{4}{7}$
 (C) $\frac{13}{56}$
 (D) $\frac{1}{14}$

4. Let $\vec{v}_1 = a^2\hat{i} - 2\hat{j} + (a+1)\hat{k}$ and $\vec{v}_2 = 2\hat{i} + \hat{j} - (a^2-1)\hat{k}$ be two vectors such that $\vec{v}_2$ lies in the plane $3x - \lambda y + z = 1$ and $(\vec{v}_1 + \vec{v}_2) \cdot \vec{v}_1 = a^4 + a^2 + 2a + 5$ then greatest volume of the parallelopiped whose co-terminous edges are represented by $\vec{v}_1, \vec{v}_2$ and $\vec{n}$ where $\vec{n}$ is the normal vector of the given plane, is -

- (A) 3
(B) 5
(C) 15
(D) 25

5. Let $A = \left\{ \frac{x^2 - p^2}{x^2 - 4x + p^2} \mid x \in \mathbb{R}, p \in \mathbb{R} \right\}$ and
 $B = \left\{ p \in \mathbb{I} \mid \frac{x^2 - p^2}{x^2 - 4x + p^2} \in \mathbb{R} \right\}$

If A is $(-\infty, \infty)$ then the value of $|B|$ is -

[Note : $|M|$ denotes the number of elements in set M]

- (A) 2
(B) 3
(C) 4
(D) 5

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. Let $y = f(x)$ satisfies the differential equation $3x^3ydx - x^4dy = (y^2 + x^3y)dx$ where $x > 0$, $y > 0$ and $x \neq \frac{1}{2}$. If $y(1) = 1$ then identify which of the following statement(s) is(are) correct ?
- (A) Area of the triangle formed by tangent and normal drawn to the curve at $(1, 1)$ with the x-axis 1 sq. unit.
- (B) If $f(x) = 1 + m(x - 1)$ has three distinct solutions then $m > 1$.
- (C) If $f(x) = 1 + m(x - 1)$ has two distinct solutions then $m \in [0, 1)$.
- (D) If $f(x) = 1 + m(x - 1)$ has two distinct solutions then $m \in (-\infty, 1)$.

7. Consider $H : \frac{x^2}{4} - \frac{y^2}{16} = 1$ and $P : y^2 = 4x$,

If $M(x, y)$ lies inside the parabola from where two distinct tangents are drawn to the hyperbola H , then identify the correct statement(s) -

- (A) If tangents touches two different branches of the hyperbola then area of the region in which $M(x, y)$ lies is $2/3$ sq. units.
- (B) If tangents touches two different branches of the hyperbola then area of the region in which $M(x, y)$ lies is $1/3$ sq. units.
- (C) If tangents touches two same branches of the hyperbola then area of the region in which $M(x, y)$ ($0 \leq x \leq 2$) lies is $\left(\frac{16\sqrt{2}-2}{3}\right)$ sq. units.
- (D) If tangents touches two same branches of the hyperbola then area of the region in which $M(x, y)$ ($0 \leq x \leq 2$) lies is $\left(\frac{8\sqrt{2}-1}{3}\right)$ sq. units

8. If $f(x) = \max(x^2 - 4, 2 - |x|)$ and $g(x) = f(f(x))$ then identify which of the following statement(s) is(are) correct ?

- (A) $g(x)$ is non-derivable at 5 points.
- (B) $\int_{-1}^2 g(x)dx = \frac{1}{2}$
- (C) The number of solution of the equation $g(x) = \frac{2015}{2014}$ is 8.
- (D) $g(x)$ is continuous for all $x \in \mathbb{R}$.

9. If maximum length of the tangent drawn to the circle which touches the straight line $x + y = 4$ at $(2, 2)$ and also the positive co-ordinate axes, from the point $(x, \pm(4 - |x|))$ where $-6 \leq x \leq 4$ is $4\sqrt{a + b\sqrt{2}}$ then $(a + b)$ is

(Where $a, b \in \mathbb{Z}$)

- (A) 11
- (B) 13
- (C) 7
- (D) 15

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 In all other cases

1. If $\int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} (\ln|3 \cos(x + \frac{\pi}{3}) + 5 \cos x| + |3 \sin^2 x - 6 \sin x \cos x + \cos^2 x - 2|) dx = \pi \ell n p + \sqrt{q}$
where $p, q \in \mathbb{Q}$ then the value of $(p \cdot q)$ is equal to -
2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x^5 + x - 2) = x^3 - 1 \forall x \in \mathbb{R}$ and g be the inverse function of f .
If $\ell = \frac{d}{dx}((x^3 + 1) \cdot g(f^3(g(f(g(x)))))) \Big|_{x=7^{1/3}}$ and $m = \frac{d^2}{dx^2} \left(\frac{f(f(g(g(f(x))))}{x^2} \right) \Big|_{x=-2}$ then the value of $(\ell(7^{-2/3}) + 8m)$ is -
3. If α is a non-real 7th root of 3 and $x = \alpha + \alpha^2$ then the value of $x^7 - 21[x^2(x+1) + x^2 + x + 1] + 15$ is -
4. Consider $f(x) = ||x - 2| - |3x - 1| + |2x + 1||$ and
 $g(x) = \left[-x^2 + \frac{3}{2}x + 1 \right]$
[Note : $[y]$ denotes greatest integer less than or equal to y .]
If m and n are the no. of points where $h(x) = f(x) \cdot g(x)$ is discontinuous and non-derivable respectively then $(m + n)$ is -
5. If $x + y + z + w = a^2 + b^2 - 4a + 4b$ where $a^2 + b^2 = 4$ and $a, b \in \mathbb{R}$ then number of quadruplets (x, y, z, w) where $x, y, z, w \in \mathbb{N}$ is -
6. Let the locus of the centre of the family of circles which intersects the parabola $x^2 - 2y + 3x = 4$ orthogonally at the point $(1/2, -9/8)$ be C and ℓ be the least distance of the curve C from $y = x - 1/x$ ($x < 0$) then $(8\sqrt{5} \ell)$ is

Note : No questions have been asked in the section III of this paper. Please leave section III blank on the OMR answer sheet.

SECTION-IV : (Maximum Marks: 16)

- This section contains **TWO (02)** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
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- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the option(s) corresponding to all the correct match(es) is (are) chosen

Zero Marks : 0 If none of the options is chosen

Negative Marks : -1 In all other cases

1.

	Column-I		Column-II
(A)	If the point $(\alpha, \alpha^2 - 3)$, $\alpha \in \mathbb{R}$ lies in acute angular region between the lines $7x - y - 15 = 0$ and $x - y + 3 = 0$ then largest integral value of α is	(P)	2
(B)	If A, B, C are three sets such that $n(A \cap B) = n(B \cap C) = n(C \cap A) = n(A \cap B \cap C) = 2$ then $n((A \times B) \cap (B \times C))$ is	(Q)	3
(C)	If the shortest distance between the lines formed by the intersection of the plane $2x - 3y + z = 1$ with the planes $x - y - z = 1$ and $x - y - z + 1 = 0$ is $2\sqrt{\frac{a}{b}}$ where a and b are prime numbers then $ b - a $ is	(R)	4
(D)	Let 3 male and 4 female teachers be ready for a ramp walk in a line, if the number of ways in which even number of female teachers are there after every male teacher in the line, is 60 (K!) then K is	(S)	5
		(T)	6

2.

	Column-I	Column-II	
(A)	Let α and β be two positive integral roots of the equation $x^3 + (221 - 4m)x^2 + 220x + 4m = 0$ where $\alpha < \beta < 20$. Then $\left(\frac{\alpha + \beta + m}{19}\right)$ is less than or equal to -	(P)	0
(B)	If the set of values of p for which the equation $\sin^{-1}(2 x - 1) = p(x - 1/2)$ has 6 distinct solutions is (a, b) then $(a + b)$ is less than or equal to -	(Q)	2
(C)	Let $\bar{x}_w$ be the weighted mean of first n natural numbers when their weights are equal to their squares respectively then $\lim_{n \rightarrow \infty} \frac{\bar{x}_w}{n}$ is less than or equal to	(R)	3
(D)	If f is polynomial such that $f(f(f(x))) - 2f(f(x)) + 3x - 7 = 0 \forall x \in \mathbb{R}$ then $\frac{1}{2} \int_0^1 (f(x) - f^{-1}(x))dx$ is less than or equal to -	(S)	5
		(T)	8

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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